

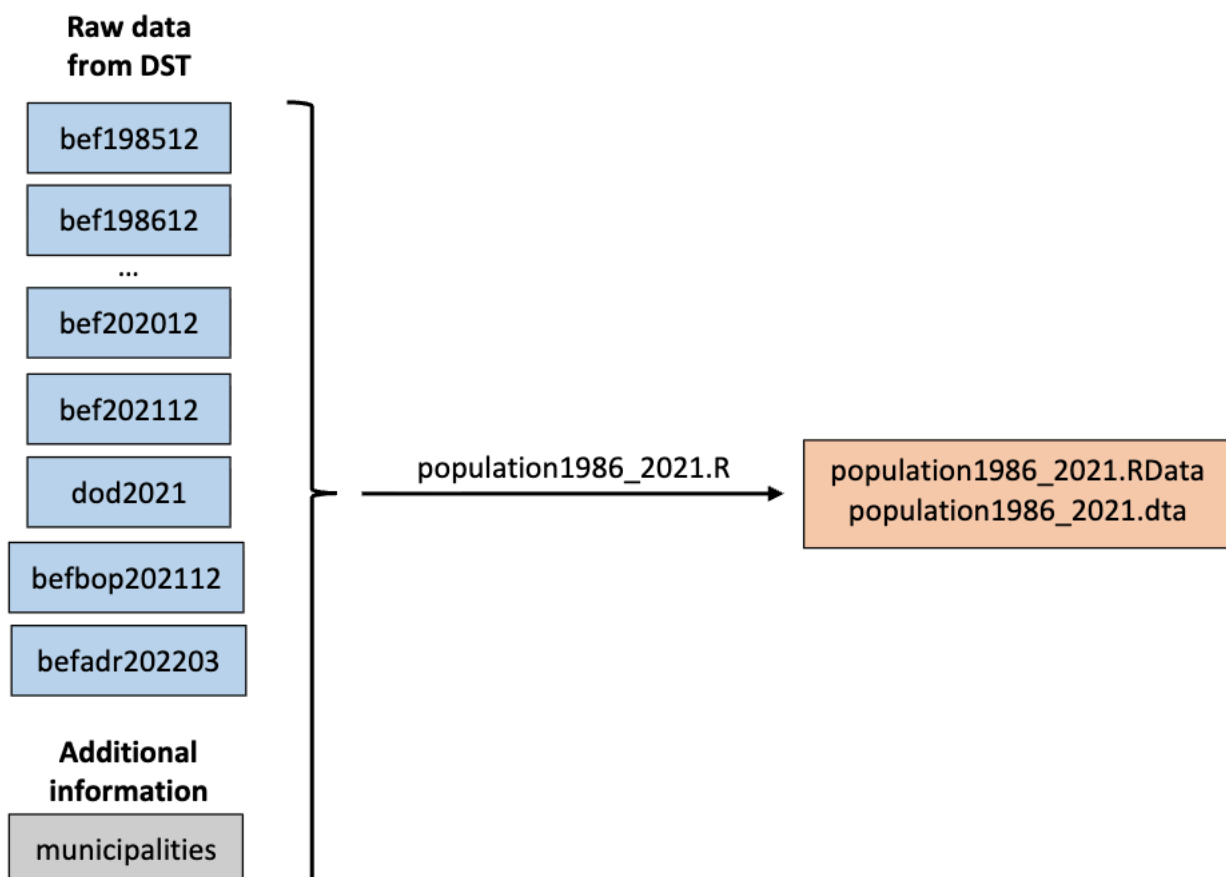
population1986_2021

Version: 1 (February 9, 2023)

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The dataset `population1986_2021` contains information on all individuals living in Denmark between January 1, 1986, and December 31, 2021. There is one observation per person with basic information (e.g., unique personal identifier, sex, date of birth, date of death, parental links).

Code: `population1986_2021.R` (R code)
Output: `population1986_2021.RData` (R file)
`population1986_2021.dta` (Stata file)



The dataset is created based on the following registers from [Statistics Denmark \(DST\)](#):

- Registers on population ([‘Befolkning’; BEF](#)), which provide information on all individuals registered in Denmark on the 31st of December of the given year. To create `population1986_2021`, we used all yearly registers from 1985 until 2021. Each BEF register is built by DST based on information from the Danish Civil Registration System,^{1–3} which was established on April 2, 1968, registering all persons alive and registered in Denmark from that point onwards.
- Register on deaths in Denmark ([‘Døde i Danmark’; DOD](#)), which provides information from the Danish Register of Causes of Death.⁴ The computerized version of the

register that we used contains the date of death for all individuals dying in Denmark and Danish citizens dying abroad from 1970 until 2021.

- Register on changes of residence ([‘Bopælsændringer’; BEFBOP](#)) from [Statistics Denmark \(DST\)](#), which contains information from the Danish Civil Registration System¹⁻³ on all individual residences in Denmark. We used information between January 1, 1986, and December 31, 2021. Additionally, we used the register on addresses ([‘Adresseregister; BEFADR](#)) and a file with Danish municipalities (municipalities.xlsx; available on <https://osf.io/zhfyp/>) to delete residences from Greenland.

To create the dataset, the population of Denmark on December 31, 2021 is identified from `bef202112` using the unique personal identifier provided by DST, and information on sex, date of birth, birthplace, origin, and unique parental identifiers is collected. Then, the population of Denmark on December, 31 of each year from 2020 to 1985 is identified iteratively from `bef202012`, `bef201912`, ..., `bef198512`, and information of those who have not already been included (from one of the more recent years) is collected. With this process, we ensure that the most recent information from each individual is used for the final dataset. Once the information on individuals registered in Denmark is collected, information on date of death is included from `dod2021` using the unique personal identifier to link the data. Finally, some individuals might be registered in the Danish Civil Registration System only for tax purposes, so we restricted the population to those with a registered address in Denmark from `befbop202112` and `befadr202203`.

The final dataset contains 8,651,587 observations from 8,651,587 unique identifiers with the following variables:

- **pnr**: Unique personal identifier. All individuals born or immigrating to Denmark receive a unique personal registration number (CPR number) containing 10 digits. Based on different digits' positions, the CPR number contains information on the date of birth and sex. The [PNR number](#) is a de-identified version of the CPR number made by DST that does not include personal information but allows linkage between different registers based on this unique identifier.
- **sex**: Sex based on DST's variable [koen](#) (1: male; 2: female).^a
- **birth_d**: Date of birth based on DST's variable [foed dag](#).
- **birth_place**: Code for the place of birth based on DST's variable [foedreg kode](#) (code for the parish or municipality for those born in Denmark or Greenland; and code for the country for those born outside Denmark or Greenland).
- **birth_country**: From `birth_place` ([foedreg kode](#)), four categories are created: Denmark, Greenland, Abroad (excluding Greenland), Unknown.
- **ie_type**: Origin with 3 categories (1: Danish origin; 2: immigrant; 3: descendent of immigrants) based on DST's variable [ie type](#).^b
- **death_d**: Date of death based on DST's variable [doddato](#).
- **pnrm**: PNR of the mother.^c
- **pnrf**: PNR of the father.^c

^a Sex indicates the legal sex of the individual based on the most recent information from the CPR number. Since September 2014, people can change their sex in the Civil Registration

System, and this data will thus sometime represent gender and not sex ([see more information](#)).

^b Origin is based on the birthplace of the index person, and citizenship and birthplace of the parents. According to DST, individuals with Danish origin include those with at least one parent who is born in Denmark and has Danish citizenship (regardless of the place of birth of the index person). Those individuals with neither of the parents being both born in Denmark and having Danish citizenship are classified as immigrants (if they are born abroad) or as descendent of immigrants (if they are born in Denmark). A descendant can change status to a person of Danish origin if at least one of their parents who is born in Denmark becomes a Danish citizen.

^c In the first years, links to parents at the Danish Civil Registration System¹⁻³ were based on families residing at the same address. From 1968 to 1978, the link was erased when either the child itself moved away from home, when a parent moved, when the child itself had children, or when the child reached age 18 years. After 1978, the links were changed from being based on addresses to being based on legal relationships (i.e., the links represent the legal parents). From 1978 onwards, links to parents were kept permanently. In 1978, an extensive revision process was initiated to establish links to parents for persons born in Denmark in 1969-1978. Today, parental links for persons born in Denmark 1969 or later are correct, and parental links for persons born in Denmark 1960-1968 are considered to be almost correct.²

1. Pedersen CB. The Danish Civil Registration System. *Scandinavian Journal of Public Health*. 2011;39(7 Suppl):22-25. doi:10.1177/1403494810387965
2. Pedersen CB, Gøtzsche H, Møller JO, Mortensen PB. The Danish Civil Registration System. A cohort of eight million persons. *Danish medical bulletin*. 2006;53(4):441-449.
3. Schmidt M, Pedersen L, Sørensen HT. The Danish Civil Registration System as a tool in epidemiology. *Eur J Epidemiol*. 2014;29(8):541-549. doi:10.1007/s10654-014-9930-3
4. Helweg-Larsen K. The Danish Register of Causes of Death. *Scandinavian Journal of Public Health*. 2011;39(7 Suppl):26-29. doi:10.1177/1403494811399958

